

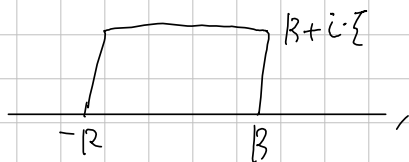
✓ Evaluate

$$\int_{-\infty}^{\infty} e^{-\pi x^2} \cdot e^{-2\pi i x \zeta} dx = e^{-\pi \zeta^2} \quad (\zeta > 0)$$

Proof. Let $f(z) = e^{-\pi z^2}$. Then, it is entire.

Consider four contour: $C_1: t \quad (-R \leq t \leq R)$,
 $C_2: R + it \quad (0 \leq t \leq \zeta)$
 $C_3: t + i\zeta \quad (-R \leq t \leq R)$
 $C_4: -R + it \quad (0 \leq t \leq \zeta)$

Then, the composition $C = C_1 \cdot C_2 \cdot (-C_3) \cdot (-C_4)$ is described as



and, by the Cauchy's theorem, $\int_C f = 0$. Now,

$$\int_{C_1} f = \int_{-R}^R e^{-\pi x^2} dx,$$

$$\int_{C_2} f = \int_0^{\zeta} e^{-\pi(R+it)^2} \cdot i dt = i \int_0^{\zeta} e^{-\pi(R^2 - t^2 + 2iRt)} \cdot dt$$

$$\text{and so } \left| \int_{C_2} f \right| = \left| \int_0^{\zeta} e^{\pi \cdot (t^2 - R^2)} dt \right| \leq \left| \int_0^{\zeta} e^{\pi t^2} dt \right| \cdot e^{-\pi R^2} \rightarrow 0 \text{ as } R \rightarrow \infty.$$

Similarly $\left| \int_{C_4} f \right| \rightarrow 0$ as $R \rightarrow \infty$. And,

$$\begin{aligned} \int_{C_3} f &= \int_{-R}^R e^{-\pi(t+i\zeta)^2} dt = \int_{-R}^R e^{-\pi(t^2 - \zeta^2 + 2it\zeta)} dt \\ &= e^{\pi \zeta^2} \cdot \int_{-R}^R e^{-\pi t^2} \cdot e^{-2\pi it\zeta} dt \end{aligned}$$

Thus, as $R \rightarrow \infty$,

$$0 = \int_{-\infty}^{\infty} e^{-\pi x^2} dx - e^{\pi \zeta^2} \int_{-\infty}^{\infty} e^{-\pi t^2} \cdot e^{-2\pi it\zeta} dt. \quad \square$$

For each $n=2,3,\dots$ evaluate

$$\int_0^{\infty} \frac{1}{1+x^n} dx.$$



Proof. Let $f(z) = \frac{1}{1+z^n}$. Take a Contour C as a Composition of:

$$C_R: t \mapsto t \quad (0 \leq t \leq R), \quad C_L: t \mapsto R \cdot e^{i \frac{2\pi}{n} t} \quad (0 \leq t \leq 2\pi),$$

$$C_S: t \mapsto t \cdot e^{i \frac{2\pi}{n}} \quad (0 \leq t \leq R), \text{ that is, } C = C_R * C_L * (-C_S).$$

Inside C , $1+z^n$ has only one zero, $\exp(i\pi/n)$, so f has only one singularity inside C , the simple pole $\exp(i\pi/n)$, by the Residue theorem,

$$\begin{aligned} \int_C f &= 2\pi i \cdot \text{Res}(f, \exp(i\pi/n)) = \lim_{z \rightarrow \exp(i\pi/n)} (z - \exp(i\pi/n)) \cdot \frac{1}{1+z^n} \cdot 2\pi i \\ &= \frac{2\pi i}{n \cdot \exp(i\pi/n)^{n-1}} = -\frac{\exp(i\pi/n)}{n} \cdot 2\pi i \end{aligned}$$

Meanwhile,

$$\begin{aligned} \left| \int_{C_L} f \right| &= \int_0^{2\pi} \left| \frac{1}{1+R^n \cdot e^{i 2\pi t/n}} \cdot \frac{R i}{n} \cdot e^{i \frac{2\pi}{n} t} \right| dt \\ &= \int_0^{2\pi} \frac{R/n}{|1+R^n \cdot e^{i 2\pi t/n}|} dt \leq \int_0^{2\pi} \frac{R/n}{|R^n - 1|} dt \rightarrow 0 \text{ as } R \rightarrow \infty. \end{aligned}$$

and

$$\begin{aligned} \int_{C_S} f &= \int_0^R \frac{e^{2\pi i t/n}}{1+t^n e^{2\pi i}} dt = e^{\frac{2\pi i}{n}} \cdot \int_0^R \frac{1}{1+t^n} dt. \text{ So that} \\ -\frac{2\pi i}{n} \cdot \exp(i\pi/n) &= \lim_{R \rightarrow \infty} \int_C f = \int_0^{\infty} \frac{1}{1+x^n} dx - \int_0^{\infty} \frac{1}{1+x^n} dx \cdot e^{\frac{2\pi i}{n}} \\ &= (1 - e^{\frac{2\pi i}{n}}) \cdot \int_0^{\infty} \frac{1}{1+x^n} dx \end{aligned}$$

$$\Rightarrow \int_0^{\infty} \frac{1}{1+x^n} dx = -\frac{2\pi i}{n} \cdot \exp(i\pi/n) / (1 - e^{\frac{2\pi i}{n}}).$$

□

Evaluate $\int_0^{\infty} \frac{x^2}{1+x^6} dx$

Proof. Let C_R be a closed curve of composition $C_R = C_x * C_r * (-C_s)$ where $C_x: t \mapsto t$ ($0 \leq t \leq R$), $C_r: t \mapsto R \cdot e^{i\pi t/6}$ ($0 \leq t \leq 2\pi$), $C_s: t \mapsto t e^{i\pi/3}$ ($0 \leq t \leq R$)

Inside the curve C , $f(z) = z^2/(1+z^6)$ has only one singularity at $z_0 = \exp(i\pi/6)$. By the Residue theorem

$$\int_{C_R} f = 2\pi i \cdot \text{Res}(f, z_0) = 2\pi i \cdot \lim_{z \rightarrow z_0} (z - z_0) \cdot \frac{z^2}{1+z^6} = z_0^2 \cdot \frac{2\pi i}{6z_0^5} = \frac{2\pi i}{6z_0^3} = \frac{\pi}{3}$$

Meanwhile, for $R > 1$,

$$\left| \int_{C_R} f \right| \leq \int_0^{2\pi} \left| \frac{R^2 e^{i\pi t/3}}{1 + R^6 e^{i\pi t}} \right| \left| \frac{\pi i \cdot R \cdot e^{i\pi t/6}}{6} \right| dt$$

$$\leq \int_0^{2\pi} \frac{R^3}{|R^6 - 1|} \cdot \frac{\pi}{6} dt \rightarrow 0 \text{ as } R \rightarrow \infty$$

$$\int_{C_s} f = \int_0^R \frac{x^2 \cdot e^{2\pi i/3}}{1+x^6} \cdot e^{\pi i/3} \cdot dt = - \int_0^R \frac{x^2}{1+x^6} dx$$

Consequently,

$$\frac{1}{3\pi} = \lim_{R \rightarrow \infty} \int_{C_R} f = 2 - \int_0^{\infty} \frac{x^2}{1+x^6} dx, \text{ so } \frac{\pi}{6} \text{ is the desired. } \square$$

Evaluate $\int_0^\infty \sin(x^2) dx$. Is it equals to $\int_0^\infty \cos(x^2) dx$?

Proof. Let $f(z) = e^{-z^2}$

Construct a Curve C as a composition $C_1 * C_2 * (-C_3)$ where

$C_1: t \mapsto t$ ($0 \leq t \leq R$), $C_2: t \mapsto Re^{it}$ ($0 \leq t \leq \frac{\pi}{4}$), $C_3: t \mapsto t \cdot e^{i\frac{\pi}{4}}$ ($0 \leq t \leq R$)

Since f is entire, $\int_C f = 0$, by the Cauchy-Goursat's theorem.

Meanwhile,

$$\left| \int_{C_2} f \right| \leq \int_0^{\frac{\pi}{4}} \left| e^{-R^2 e^{2it}} \cdot R i \cdot e^{it} \right| dt = \int_0^{\frac{\pi}{4}} e^{-R^2 \cos 2t} \cdot R dt$$

Let $u = \pi/4 - t$, then $\cos 2t = \cos(\frac{\pi}{2} - 2u) = \sin 2u$

$$\begin{aligned} \text{Then, } \int_0^{\frac{\pi}{4}} e^{-R^2 \cos 2t} \cdot R dt &= \int_0^{\frac{\pi}{4}} e^{-R^2 \sin 2u} \cdot R du \\ &\leq R \cdot \int_0^{\frac{\pi}{4}} e^{-R^2 \cdot 4u/\pi} du \\ &= R \cdot \frac{\pi}{4R^2} \cdot (1 - e^{-R^2}) \rightarrow 0 \text{ as } R \rightarrow \infty. \end{aligned}$$

Therefore, $\lim_{R \rightarrow \infty} \int_{C_2} f = 0$. And,

$$\begin{aligned} \int_{C_3} f &= \int_0^R e^{-t^2} e^{i\pi/2} \cdot e^{i\frac{\pi}{4}} dt = \int_0^R e^{-it^2} \cdot e^{i\pi/4} dt \\ &= e^{i\pi/4} \cdot \int_0^R \cos(t^2) - i \sin(t^2) dt \\ &= \frac{\sqrt{2}}{2} \cdot (1 + i) \cdot \int_0^R \cos(t^2) - i \sin(t^2) dt \\ &= \frac{\sqrt{2}}{2} \cdot \left(\int_0^R \cos(t^2) + \sin(t^2) dt + i \int_0^R \cos(t^2) - \sin(t^2) dt \right) \end{aligned}$$

Now,

$$\lim_{R \rightarrow \infty} \int_C f = \int_0^\infty e^{-x^2} dx + 0 - \frac{\sqrt{2}}{2} \cdot \left(\int_0^\infty \cos t^2 + \sin t^2 dt + i \int_0^\infty \cos t^2 - \sin t^2 dt \right) = 0$$

So that the imaginary part is zero, thus $\int_0^\infty \cos(t^2) - \sin(t^2) dt = 0$, and

$$\int_0^\infty \cos(x^2) dx = \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{2}}{4}. \quad \square$$

Evaluate $\int_0^{\infty} \frac{\sin x}{x} dx$ using indented semicircle.

Proof. Let $f(z) = (e^{iz} - 1)/z$, and let C be a contour of composition

$C_1: t \mapsto \varepsilon \cdot e^{it} \ (0 \leq t \leq \pi)$, $C_2: t \mapsto t \ (\varepsilon \leq t \leq R)$,

$C_3: t \mapsto R e^{it} \ (0 \leq t \leq \pi)$, $C_4: t \mapsto t \ (-R \leq t \leq -\varepsilon)$, $C = [-C_1] * C_2 * C_3 * C_4$.

By the Cauchy-Goursat theorem, $\int_C f(z) dz = 0$.

Meanwhile,

$$\int_{C_1} f = \int_0^{\pi} \frac{e^{i\varepsilon e^{it}} - 1}{\varepsilon \cdot e^{it}} \cdot \varepsilon \cdot i \cdot e^{it} dt = \int_0^{\pi} e^{-\varepsilon(\sin t - i \cos t)} - 1 dt$$

By dominant convergence theorem,

$$\lim_{\varepsilon \rightarrow 0} \int_{C_1} f = \int_0^{\pi} \lim_{\varepsilon \rightarrow 0} e^{-\varepsilon(\sin t - i \cos t)} - 1 dt = 0$$

✓ Evaluate $\int_0^{\infty} \frac{\log x}{1+x^4} dx$